

Deep Networks

MLP/FNN to Universal Function Approximation, Depth and Width Efficiency

Introduction to Deep Learning · Class 9

10 August

Outline

The Multilayer Perceptron, Formally

Universal Approximation, Revisited

How Wide Must a Network Be?

How Much Does Depth Buy?

Counting Capacity

Reading the Theorems

The Multilayer Perceptron, Formally

Notation used throughout this course

$\mathbf{x} \in \mathbb{R}^D$	input / features	$\phi_j(\mathbf{x})$	fixed basis function
$y, \hat{y}$	target, model prediction	$\mathcal{K} \subset \mathbb{R}^D$	a compact set
$\mathbf{z}^{(l)} = h(\mathbf{a}^{(l)})$	units of hidden layer l	$\mathcal{C}(\mathcal{K})$	continuous functions on $\mathcal{K}$
$\mathbf{W}^{(l)}, \mathbf{b}^{(l)}$	weights and biases	$\ f\ _\infty$	$\sup_{\mathbf{x} \in \mathcal{K}} f(\mathbf{x}) $
L, M	depth, width	ε	target accuracy
W	total number of parameters	$\mathcal{N}_{L,M}$	functions a network can compute

Symbols follow **Bishop & Bishop (2024)**. W counts every weight and bias; $\mathcal{N}_{L,M}$ is the set of functions a network of that shape can compute.

Where Class 7 left us

- Composition **multiplies** linear pieces; width only **adds** them
- Universal approximation holds in D dimensions for any non-polynomial activation, via ridge functions
- ReLU networks compute **exactly** the continuous piecewise-linear functions — shallow or deep
- The sawtooth needs $O(L)$ units deep and $2^L - 1$ shallow

All of that was about **what** a network can represent. None of it says **how much network** it takes.

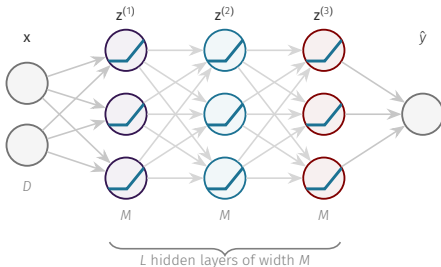
Three questions, and where they are answered

	Question	Answered by	
1	Can one hidden layer reach any function at all?	Cybenko, Hornik, Leshno	§2
2	How fast does the error fall as the network grows?	Barron; DeVore et al.	§2
3	How much of the budget should be width, and how much depth?	Lu et al.; Telgarsky; Eldan & Shamir; Yarotsky	§3–4

A fourth question follows from the third: if depth buys so much expressive power, what does that power **cost** in data? That is §5.

The multilayer perceptron

Before we can count anything, we need to fix exactly what we are counting.



$$\mathbf{z}^{(0)} = \mathbf{x}, \quad \mathbf{z}^{(l)} = h(\mathbf{W}^{(l)}\mathbf{z}^{(l-1)} + \mathbf{b}^{(l)}), \quad \hat{\mathbf{y}} = \mathbf{W}^{(L+1)}\mathbf{z}^{(L)} + \mathbf{b}^{(L+1)}$$

Write $\mathcal{N}_{L,M}$ for the set of functions this computes as the weights range over $\mathbb{R}^W$, with

$$W = M(D+1) + (L-1)M(M+1) + (M+1) \text{ parameters.}$$

Why not just use fixed basis functions?

Before calling learned features progress, see what goes wrong without them.

Tile the input space, one basis function per cell (B&B, 2024, §6.1):

$$\hat{y}(\mathbf{x}) = \sum_j w_j \phi_j(\mathbf{x}), \quad \phi_j \text{ centred on cell } j$$

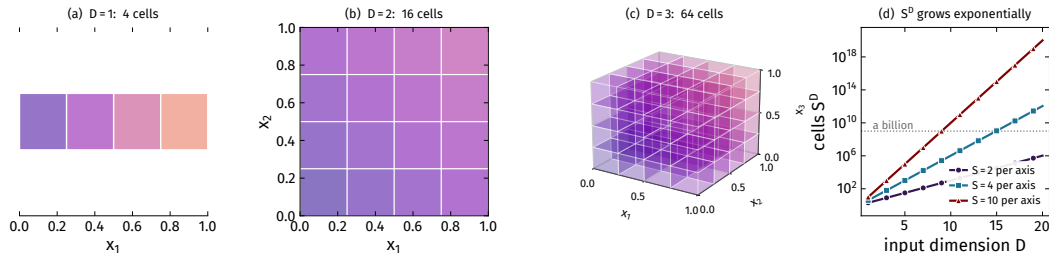
Proposition 1 (The grid blows up)

Partition each of the D axes into S intervals. The tensor-product grid then has S^D cells, so the model has S^D parameters — exponential in the input dimension.

Proof. A cell is one interval chosen per axis, so there are $|\{1, \dots, S\}^D| = S^D$. □

With $S = 10$ and a 32×32 greyscale image ($D = 1024$): 10^{1024} cells.

The curse of dimensionality, drawn

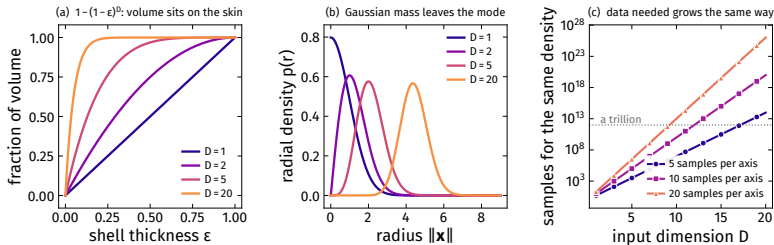


(a–c) a grid with $S = 4$ cells per axis in one, two and three dimensions: 4, 16, 64 cells. (d) the count S^D against dimension, at three resolutions.

Every cell needs its own parameter — and its own data to fit it.

Two more faces of the same problem

High-dimensional space is strange in two further ways that matter later.



(a) the volume within ϵ of a unit ball's surface, $1 - (1 - \epsilon)^D$. (b) a Gaussian's mass migrates to a shell at radius $\approx \sqrt{D}$. (c) matching a sample density costs exponentially more data.

SHALLOW fixed ϕ_j : one parameter per cell, S^D of them

DEEP learned features: units *move* to where the data is

Universal Approximation, Revisited

What “universal” means

Networks are supposed to escape this. First, say precisely what that claim asserts.

Fix a compact $\mathcal{K} \subset \mathbb{R}^D$ and use the supremum norm $\|f\|_\infty = \sup_{\mathbf{x} \in \mathcal{K}} |f(\mathbf{x})|$.

Definition

A family $\mathcal{F} \subset C(\mathcal{K})$ is **universal** (dense) if for every $f \in C(\mathcal{K})$ and every $\varepsilon > 0$ there is $g \in \mathcal{F}$ with $\|f - g\|_\infty < \varepsilon$.

- Note what is quantified: $\mathcal{F}$ is the family with **unbounded** width, so “ g exists” may mean a network of astronomical size
- Density says nothing about **how fast** the error falls as the network grows — that needs a **rate**

The classical statement

With the definition in place, the theorem is short.

Theorem 2 (Universal approximation; Cybenko 1989, Hornik 1991, Leshno et al. 1993)

Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be continuous. The family

$$\mathcal{N}_{1,\infty} = \left\{ \mathbf{x} \mapsto \sum_{j=1}^M w_j^{(2)} h(\mathbf{w}_j^{(1)\top} \mathbf{x} + w_{j0}^{(1)}) + w_0^{(2)} : M \in \mathbb{N} \right\}$$

is dense in $C(\mathcal{K})$ for every compact $\mathcal{K}$ if and only if h is not a polynomial.

- Proved in Class 7 by reduction to one dimension through **ridge functions**
- **One hidden layer is enough** — so universality alone can never be an argument for depth

Two objects the next two theorems need

Density says the error reaches zero. To ask how fast, we must first say which functions we mean.

The Barron norm

$$C_f = \int \|\omega\| |\tilde{f}(\omega)| d\omega$$

$\tilde{f}(\omega)$ is the Fourier transform: how much of f sits at frequency ω . Weighting by $\|\omega\|$ **charges a function** for its high-frequency content.

A smoothness ball

$$\mathcal{F} = \{f: \|\partial^\alpha f\|_\infty \leq 1 \ \forall |\alpha| \leq s\}$$

Every derivative up to order s is bounded by one.
This is the Sobolev ball $W^{s,\infty}$: **all functions of a** given smoothness, taken together.

One measures a **single** function's wiggleness, the other a **whole class** — which is why the two bounds point opposite ways. A bound over a ball is worst-case.

A rate that does not see the dimension

Take a function with small C_f . How quickly does a one-hidden-layer network close in on it?

Theorem 3 (Barron, 1993)

Let $f: \mathcal{K} \rightarrow \mathbb{R}$ have finite Barron norm $C_f < \infty$. Then for every M there is a one-hidden-layer sigmoidal network $\hat{y}_M$ with M units and

$$\int_{\mathcal{K}} (f(\mathbf{x}) - \hat{y}_M(\mathbf{x}))^2 \mu(d\mathbf{x}) \leq \frac{(2C_f r)^2}{M},$$

with r the radius of $\mathcal{K}$. The bound is *independent of D* .

- **Proof sketch.** Write f as a continuous mixture of ridge functions, with total mass controlled by C_f
- So f is in the closed convex hull of scaled ridge functions of norm $\leq 2C_f r$
- Maurey's lemma: a point of the convex hull of norm- b elements lies within $b/\sqrt{M}$ of an average of M of them
- That average is an M -unit network

□

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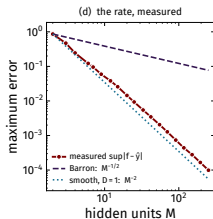
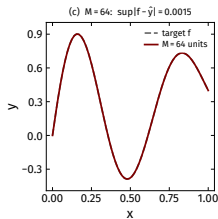
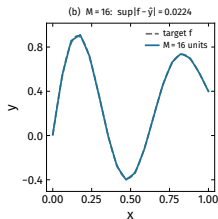
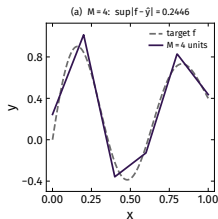
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□

The rate, measured



- (a–c) the best M -unit fit to one smooth target as M grows
- (d) the measured error against two references
- Barron's $M^{-1/2}$ is a **worst-case guarantee** over a whole class; a single smooth target in $D = 1$ does much better, here M^{-2}
- A theorem gives a ceiling on the error, not a prediction of it

And a lower bound that does

Barron's class is generous to us. What happens if we ask for a guarantee over all functions of a given smoothness instead?

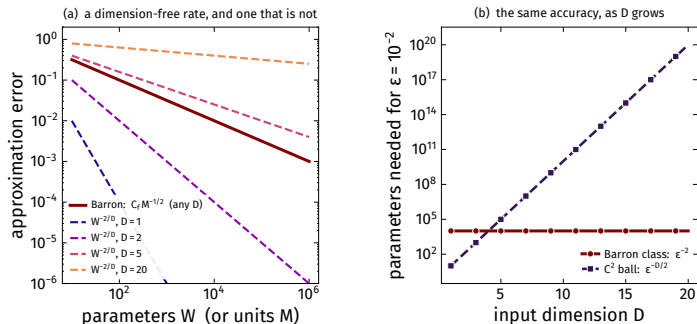
Theorem 4 (DeVore, Howard & Micchelli, 1989)

*Let $\mathcal{F}$ be the unit ball of the Sobolev space $W^{s,\infty}([0,1]^D)$. If an approximation scheme uses W parameters chosen *continuously* from the target, then its worst-case error over $\mathcal{F}$ is at least $cW^{-s/D}$.*

- Smoothness s buys accuracy; dimension D takes it away. To halve the error in D dimensions you need $2^{D/s}$ times the parameters
- It limits *every* scheme with continuous parameter selection, not networks in particular
- No contradiction with Barron: his class is *much smaller* than a Sobolev ball

The curse is real. It is escaped only by targets with extra structure — which is exactly what real data has.

The two bounds side by side



(a) the dimension-free Barron rate against the Sobolev lower bound for several D : for $D = 1$ or 2 smoothness wins, but by $D = 20$ the Barron bound is far better. (b) parameters needed for $\epsilon = 10^{-2}$: constant in the Barron class, exponential in D on a C^2 ball.

How Wide Must a
Network Be?

Width as a threshold: the upper bound

We know the cost of accuracy. Now: how to spend the budget. Width behaves quite unlike depth.

Theorem 5 (Width-bounded universality; Lu, Pu, Wang, Hu & Wang, 2017)

*ReLU networks of width $D + 4$ and **arbitrary depth** are dense in $L^1(\mathbb{R}^D)$: for every Lebesgue-integrable f and every $\varepsilon > 0$ there is such a network with $\int |f - \hat{y}| \, d\mathbf{x} < \varepsilon$.*

- Four units more than the input dimension is **all** the width that is ever needed, provided the network may be deep
- Note the norm: density in L^1 , not uniform approximation

Width as a threshold: the lower bound

Four units of slack sounds arbitrary. It is not — just below it, everything fails.

Theorem 6 (Width lower bound; Hanin & Sellke 2017, Johnson 2019)

ReLU networks of width $\leq D$ are not universal, however deep. There are continuous functions they cannot approximate to better than a fixed constant.

Why it holds. A ReLU network of width $\leq D$ never expands the space it maps through, so all of its **level sets are unbounded**. A function with a bounded level set — say $\|x\|$, whose level sets are spheres — cannot be approximated.

Width is a threshold: below D nothing works, just above it everything does. **Depth is a resource:** more of it buys efficiency.

How Much Does Depth
Buy?

The witness, recalled

Width has a floor and then stops mattering. Depth has no such ceiling.

Lemma 7 (The sawtooth, from Class 7)

$T(x) = 2h(x) - 4h(x - \frac{1}{2})$ is a two-unit ReLU network that maps $[0, 1]$ onto $[0, 1]$, folding it in half. Its L -fold composition $T^{\circ L}$ has exactly 2^L linear pieces and is computed by a network of depth L and width 2.

Two units per layer, L layers — and the number of pieces **doubles with every layer added**. That is the whole mechanism.

Depth beats width, provably

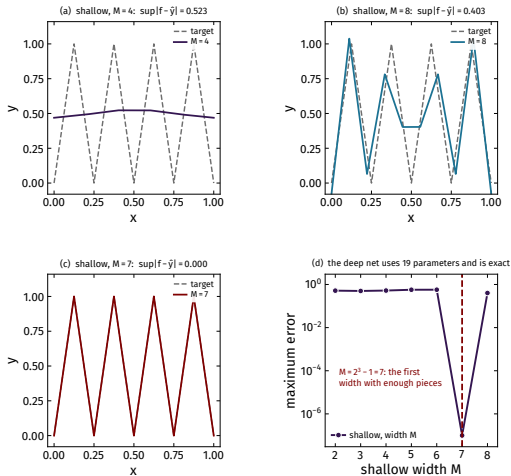
Theorem 8 (Depth–width separation; Telgarsky, 2016)

$T^{\circ L}$ is computed by a ReLU network with $O(L)$ parameters, but any *one-hidden-layer* ReLU network computing it exactly needs at least $2^L - 1$ units. Telgarsky's full result strengthens “exactly” to “to within a constant”.

Proof of the lower bound. A one-hidden-layer ReLU network with M units on a scalar input is piecewise linear with at most $M + 1$ pieces. Matching 2^L pieces forces $M + 1 \geq 2^L$. $\square$

An *exponential* gap, from a function you can draw in one line.

The separation, measured



A separation that is not one-dimensional

The sawtooth lives on a line — easy to dismiss. The same gap appears in D dimensions.

Theorem 9 (Eldan & Shamir, 2016)

*There is a **radial** function $g : \mathbb{R}^D \rightarrow \mathbb{R}$, supported on a ball, that a network with **three** layers and $\text{poly}(D)$ width approximates to within a constant in L^2 , but which **every two-layer** network needs width $\exp(\Omega(D))$ to approximate that well.*

- Why radial functions are hard for two layers: a two-layer net is a sum of **ridge** functions, each constant along $D - 1$ directions, and building a sphere out of hyperplanes is expensive
- Three layers can first compute $\|\mathbf{x}\|^2 = \sum_i x_i^2$ with D cheap units, then apply a one-dimensional function to it
- Unlike Telgarsky's, this separation is about **approximation in L^2** , not exact representation, and grows with the **dimension** rather than the depth

How deep networks beat the grid

Separations show depth can win. This says what it costs.

Theorem 10 (Yarotsky, 2017)

For f in the unit ball of $C^n([0, 1]^D)$ and any $\varepsilon \in (0, 1)$ there is a ReLU network with

$$\text{depth } O(\log(1/\varepsilon)) \quad \text{and} \quad W = O(\varepsilon^{-D/n} \log(1/\varepsilon)) \text{ parameters}$$

*such that $\|f - \hat{y}\|_\infty \leq \varepsilon$. Up to the logarithm this is **optimal**.*

- Deep ReLU networks approximate x^2 , hence products, with $O(\log(1/\varepsilon))$ layers — the sawtooth again

Depth removes the **overhead**, not the curse.

Counting Capacity

Counting linear regions

Rates are one currency; counting the pieces is another, and easier to compute.

Theorem 11 (Lower bound; Montúfar, Pascanu, Cho & Bengio, 2014)

A ReLU network with L hidden layers of width $M \geq D$ on $\mathbb{R}^D$ can realise at least

$$\left(\prod_{l=1}^{L-1} \lfloor M/D \rfloor^D \right) \sum_{j=0}^D \binom{M}{j}$$

linear regions.

The trailing $\sum_{j \leq D} \binom{M}{j}$ is the **Zaslavsky count** from Class 5: the last layer behaves like a shallow network, and the product in front is what depth adds.

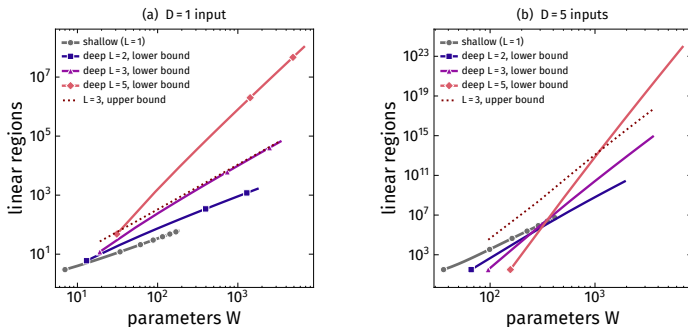
Counting linear regions: the ceiling

Theorem 12 (Upper bound; Serra, Tjandraatmadja & Ramalingam, 2018)

The number of regions is at most $\prod_{l=1}^L \sum_{j=0}^D \binom{M}{j}$, and tighter bounds follow from the dimensions of the individual layers.

- Both bounds are **exponential in L** and only **polynomial in M** — the same asymmetry as the sawtooth
- A lower bound says such networks **exist**; it does not say training finds them

Regions against parameters



Regions against the parameter count W , for (a) a scalar input and (b) $D = 5$. Grey is the shallow count; the coloured curves are the Montúfar lower bound at three depths; dotted is the product upper bound at $L = 3$.

Widths are taken in multiples of D , where the lower bound is stated.

Capacity has a price

Every result so far has been an argument for depth. Here is the bill.

Theorem 13 (VC dimension; Bartlett, Harvey, Liaw & Mehrabian, 2019)

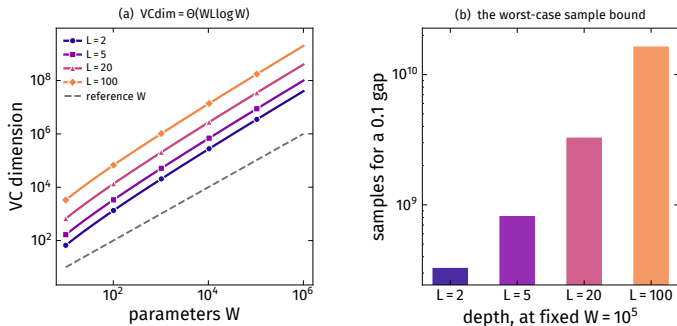
For ReLU networks with W parameters and L layers,

$$\text{VCdim} = \Theta(WL \log W).$$

- Learning theory then needs $N = \Omega(\text{VCdim}/\epsilon^2)$ samples for a gap ϵ
- Modern networks have $\text{VCdim} \gg N$ and generalise anyway: the bound is **loose** for real data (Class 17)

Expressivity and generalisation theorems pull in **opposite directions**; neither alone explains deep learning.

VC dimension, drawn



(a) $\Theta(WL \log W)$ against the parameter count, at four depths, with a linear reference. (b) at a fixed $W = 10^5$, the worst-case number of samples the VC bound demands for a generalisation gap of 0.1 — growing linearly with depth.

Reading the Theorems

What each theorem actually gives

Ten results in one lecture. What each one is really claiming:

Result	Gives	Does not give
Cybenko / Hornik / Leshno	density with one hidden layer, any non-polynomial h	any bound on the width
Barron (1993)	$O(C_f/\sqrt{M})$, free of D	anything outside the Barron class; C_f may grow with D
DeVore et al. (1989)	$W^{-s/D}$ is unbeatable on a Sobolev ball	a statement about one particular target
Lu et al. (2017)	width $D+4$ suffices, given depth	efficiency at that width
Telgarsky (2016)	an explicit exponential depth gap	that such functions matter in practice
Yarotsky (2017)	$O(\varepsilon^{-D/n} \log \frac{1}{\varepsilon})$, depth $O(\log \frac{1}{\varepsilon})$	escape from the curse
Montúfar (2014)	regions exponential in L	that training reaches them
Bartlett et al. (2019)	$\text{VCdim} = \Theta(WL \log W)$	a useful bound at modern scale

What none of them say

And the gap between all this and a working network:

- **Nothing about optimisation.** Every result is about the *existence* of weights; finding them is Classes 11–14
- **Nothing about generalisation.** The capacity bounds are worst-case over all data distributions
- **The hard cases may be irrelevant.** The sawtooth and the radial bump are built to be hard; natural signals are not adversarial
- **Architecture matters more than the counts suggest.** Convolution and attention change the function class itself

SHALLOW universal, but with a vacuous width bound

DEEP universal *and* efficient, on the functions we care about

Takeaways

1. Fixed basis functions need S^D parameters. Networks avoid this by **placing their features**, not tiling the space
2. One hidden layer is already universal, for any non-polynomial activation — so **universality alone never argues for depth**
3. Rates are what matter. **Barron**: $O(C_f/\sqrt{M})$, free of dimension, on a restricted class. **DeVore et al.**: $W^{-s/D}$ is unbeatable on a smoothness ball
4. **Width is a threshold** ($\leq D$ fails, $D + 4$ suffices); **depth is a resource** — Telgarsky and Eldan–Shamir give explicit exponential gaps, and Yarotsky shows depth $O(\log \frac{1}{\epsilon})$ removes the overhead
5. Capacity counts both ways: regions grow exponentially in L , and so does the worst-case **VC bound**

Class 11: having established what these networks can represent, how do we actually **train** them?

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- P. Bartlett, N. Harvey, C. Liaw & A. Mehrabian, *JMLR* 20:63 (2019) — VC dimension.
- All plots generated by `src/build_all.py`; one module per figure under `src/figures/`.

Questions?